Jinwei Gu

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EDUCATION:

9/2005-5/2010	**Columbia University** Ph.D. in Computer Science (Honor: PhD with Distinction)
9/2002-5/2005	**Tsinghua University** M.S. in Automation (Honor: Outstanding Master's Thesis Award)
9/1998-5/2002	**Tsinghua University** B.S. in Automation

EMPLOYMENT:

05/2024-Present	NVIDIA	Principal Research Scientist
03/2023-Present	The Chinese University of Hong Kong (CUHK)	Adjunct Associate Professor (Department of Computer Science & Engineering)
11/2018-03/2023	SenseBrain (SenseTime USA)	R&D Executive Director
11/2015-11/2018	NVIDIA	Senior Research Scientist
11/2013-10/2015	Futurewei Technologies	Senior Research Scientist
9/2010-6/2013	Rochester Institute of Technology (RIT)	Assistant Professor of the Chester F. Carlson Center for Imaging Science
9/2005-5/2010	Columbia University	Graduate Research Assistant of the CAVE Laboratory in Computer Science
5/2008-8/2008	Adobe Research	Research Intern in the CTL Lab
2003-2004	Microsoft Research Asia (MSRA)	Research Intern in the Visual Computing Lab

RESEARCH INTERESTS:

Computational Imaging; AI Image Sensor/Camera System; 3D Computer Vision; Computational Photography; Computer Graphics; Machine Learning

RESEARCH PRODUCT EXPERIENCE:

NVIDIA (2024-present)
One of the tech leads and core contributors for the development and CES 2025 release for NVIDIA Cosmos World Foundation Models (which won the Best of CES Award), including building Cosmos Tokenizers, post-training of world foundation models for robotics applications, implementing automatic benchmarking system for evaluations of video foundation models, and the development of Cosmos-Transfer models.

One of the core contributors for NVIDIA Cosmos Transfer1, a Multimodal ControlNet based on Cosmos video foundation model. Cosmos Transfer is used together with physics simulation engines (e.g., NVIDIA Omniverse) for data generation for physical AI.

One of the contributors for NVIDIA Cosmos Reason1, a suite of models, ontologies, and benchmarks that we develop with the goal of enabling multimodal LLMs to generate physically grounded responses.

Chinese University of Hong Kong: Computational Photography with Generative AI (2023-present)
Conduct research on novel image capturing, rendering, and editing with Generative AI and LLMs, including VQGAN-based class-agnostic image restoration, text-based flexible image enhancement/editing system, closed-loop image quality assessment and enhancement, NeRF, inverse rendering, etc.

Key tech lead contributions across the full Cosmos journey: Cosmos Tokenizer (neural image/video tokenizer suite, 2048x compression, 12x faster than SOTA); Cosmos World Foundation Model / Cosmos1 (Jan 2025, Best of CES 2025); Cosmos-Transfer1 (Sim2Real multimodal ControlNet, Mar 2025); Cosmos-Reason1 (embodied reasoning suite with LLMs, Mar 2025); Cosmos World Simulation / Cosmos2.5 (Oct 2025, flow-based architecture trained on 200M video clips); Cosmos-Transfer2.5 (Sim2Real/Real2Real translation); Cosmos-Policy (ICLR 2026, fine-tuning video models for robot policy); as well as derived Physical AI applications: Scalable Policy Evaluation, DreamDojo (robot world model from human videos), and Plenoptic Video Generation.
SenseBrain 2018-2023: Computational Photography for Smart Phones
Build and lead a R&D team of ~30 people in Silicon Valley, focusing on complete product solution for photo/video quality enhancement, which is computational photography and computational imaging on mobile phones, including Super-Resolution, Smart Image/Video Denoising, HDR Video, AI Camera ISP, Portrait Enhancement, RGBW image sensor, RGBCMY image sensor, Super Night, Under Display Camera, etc. (with close collaboration with Tetras.AI & SONY). In 2020, we built the world's first

AI-sensor (RGBW, IMX866) and the first 200M-pixel camera system (2*2*4OCL, IMX777) and commercialized them into flagship phones with Tetras.AI & SONY.

NVIDIA 2017-2018: Co-Pilot SDK / Drive IX (AI-based ADAS Platform)
Key member in the team for technology development and transfer to build the Co-Pilot SDK (AI for in-car monitoring), including head pose estimation, facial key point tracking, gaze estimation and tracking, and hand gesture recognition. Demo in NVIDIA Keynote for CES 2017, Jan 2017
https://techcrunch.com/2017/01/04/nvidia-builds-a-co-pilot-into-its-autonomous-drive-computer/ Our work on wide-baseline video stitching is also shipped into NVIDIA Drive IX SDK (https://www.nvidia.com/en-us/self-driving-cars/drive-ix/). I also led the team developing novel approaches for DNN-based inverse image rendering, video-based depth estimation, 3D plane estimation, polarization-based multiview stereo, etc.

NVIDIA 2015-2017: VirtualEye Project (multi-camera tracking + free-view video)
Key member in this Nvidia+DARPA collaboration, June 2016
https:/www.engadget.com/2016/06/24/darpas-virtual-eye-lets-soldiers-see-around-obstacles We built a multi-camera tracking system to create free-view video experiences for telecommunication and telepresence. The entire system includes: multi-camera Co-SLAM, 3D reconstruction of background and foreground subjects, novel view synthesis, and real-time streaming for image-based rendering. I am in charge of the multi-camera Co-SLAM tracking and the system design, and I also ported the Co-SLAM system on NVIDIA Jetson TX2 for mobile devices, with lower power consumption.

Futurewei 2014-2015: V-Sports Project (multi-camera virtual reality system)
Team lead for prototyping a complete virtual reality system built with multiple cameras. The system includes: multi-camera 3D scene capture and post-processing (i.e., 3D alignment, background subtraction, etc.), multi-camera video stitching, real-time streaming, and augmented reality on mobile devices (Android).

RIT 2012-2013: Discriminative Illumination for Recycling (material classification)
Principal investigator for building a novel active lighting and imaging system for recycling scrap materials (e.g., aluminum, copper, plastics). I led the team to build a LED-based multispectral dome (controlled with Arduino boards) as a prototype product and test with real samples from the institute of sustainability for scrap material recycling.

HONORS:

Best of CES Award (2025) — NVIDIA Cosmos World Foundation Models
CVPR 2019 Best Paper Final List
CVPR 2018 Outstanding Reviewer
Ph.D. Degree with Distinction, Columbia University, 2010.
Ph.D. Service Award, Columbia University, 2009.
Principal Scholarship Award, Tsinghua University, 2005.
Outstanding M.S. Thesis Award, Tsinghua University, 2005.

PUBLICATIONS[1]*: (to be updated)

[1] In computer vision, one of the leading journals is the IEEE Transactions on Pattern Analysis and Machine

For latest publication, please use Google Scholar Link: https://scholar.google.com/citations?user=k_T8t30AAAAJ&hl=en&oi=ao

NVIDIA. Cosmos-Transfer1: Conditional World Generation with Adaptive Multimodal Control. Arxiv. 2503.14492. 2025.

NVIDIA. Cosmos-Reason 1: From Physical AI Common Sense to Embodied Decisions. Arxiv.

NVIDIA. Cosmos World Foundation Model Platform for Physical AI. Arxiv. 2501.03575. 2025

Refereed journal articles:

J19. H. Wu, C. Chen, L. Liao, J. Hou, W. Sun, Q. Yan, J. Gu, W. Lin. "Neighbourhood representative sampling for efficient end-to-end video quality assessment", IEEE TPAMI, volume 45, issue 12, 2023.

J18. Kevin Chan, Xiangyu Xu, Xintao Wang, **Jinwei Gu**, and Chen Change Loy. "GLEAN: Generative Latent Bank for Image Super-Resolution and Beyond", *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)*, 45(3), 2023.

J17. Josh Rego, Huaijin Chen, Shuai Li, **Jinwei Gu**, and Suren Jayasuriya. "Deep Camera Obscura: An Image Restoration Pipeline for Pinhole Photography", *Optics Express* 30 (15), 27214-27235, 2022.

J16. Shiyu Duan, Huaijin Chen, **Jinwei Gu**. "JPD-SE: High-level Semantics for Joint Perception-Distortion Enhancement in Image Compression.", *IEEE Transactions on Image Processing, vol. 31, 2022*.

J15. Chongyi Li, Chunle Guo, Linghao Han, Jun Jiang, Ming-Ming Chen, **Jinwei Gu**, and Chen Change Loy. "Low-light Image and Video Enhancement Using Deep Learning: A Survey", *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), vol. 44,* 2022 .

J14. Felipe Gutierrez-Barragan, Huaijin Chen, Mohit Gupta, Andreas Velten, and **Jinwei Gu**. "iToF2dToF: A Robust and Flexible Representation for Data-Driven Time-of-Flight Imaging". *IEEE Transactions on Computational Imaging*, vol.7, 2021.

Intelligence (PAMI), Leading conferences are IEEE CVPR, ICCV and ECCV. Papers in these conferences are rigorously reviewed, with acceptance rates around 20%, and below 5% for oral presentations. In computer graphics, the SIGGRAPH conferences (SIGGRAPH & SIGGRAPH Asia) are the leading venue for publishing research in computer graphics, with acceptance rates ranging from 15%-20%. Since 2002, the proceedings have also been published as a special issue of the ACM Transactions on Graphics (TOG), the leading journal in graphics. The EuroGraphics Symposium on Rendering (EGSR) is the leading venue for rendering research (second only to the SIGGRAPH conference), and has a competitive acceptance rate of 25-35%. In computational photography, the leading conference is IEEE ICCP, with the acceptance rates around 20%.

* Co-authors are my advisee (graduate students, interns or visiting scholars).

J13. Chao Liu* and **Jinwei Gu**. "Discriminative Illumination: Per-Pixel Classification of Raw Materials based on Optimal Projections of Spectral BRDF", *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 36(1), pp.86-98, 2014.*

J12. Dengyu Liu*, **Jinwei Gu**, Yasunobu Hitomi, Mohit Gupta, Tomoo Mitsunaga, and Shree Nayar. "Efficient Space-Time Sampling with Pixel-wise Coded Exposure for High Speed Imaging", *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 36(2), pp.248-260, 2014.*

J11. **Jinwei Gu**, Shree Nayar, Eitan Grinspun, Peter Belhumeur, and Ravi Ramamoorthi. "Compressive Structured Light for Recovering Inhomogeneous Participating Media," *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 35(3), pp.555-567,* 2013

J10. **Jinwei Gu**, Ravi Ramamoorthi, Peter Belhumeur and Shree Nayar. "Removing Image Artifacts Due to Dirty Camera Lenses and Thin Occluders." *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia),* Dec. 2009.

J9. Wojciech Matusik, Boris Ajdin, **Jinwei Gu**, Jason Lawrence, Hendrik P.A. Lensch, Fabio Pellacini, and Szymon Rusinkiewicz. "Printing Spatially-Varying Reflectance". *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia),* Dec. 2009.

J8. Jie Zhou, Fanglin Chen, and **Jinwei Gu**. "A Novel Algorithm for Detecting Singular Points from Fingerprint Images". *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI),* vol.31, no. 7, 2009.

J7. **Jinwei Gu**, Chien-I Tu, Ravi Ramamoorthi, Peter Belhumeur, Wojciech Matusik, and Shree Nayar. "Time-Varying Surface Appearance: Acquisition, Modeling, and Rendering". *ACM Transactions on Graphics (Proceedings of SIGGRAPH),* vol. 25, no. 3, June 2006, pp 762-771.

J6. **Jinwei Gu**, Jie Zhou, and Chunyu Yang. "Fingerprint Recognition by Combining Global Structure and Local Cues". *IEEE Transactions on Image Processing*, vol. 15, no. 7, 2006, pp 1952-1964.

J5. Wei Jiang, Guihua Er, Qionghai Dai, and **Jinwei Gu**. "Similarity-based Online Feature Selection in Content-based Image Retrieval". *IEEE Transactions on Image Processing*, vol. 15, no. 3, 2006, pp 702-712.

J4. Wei Jiang, Guihua Er, Qionghai Dai, and **Jinwei Gu**. "Hidden Annotation for Image Retrieval with Long-term Relevance Feedback Learning". *Pattern Recognition*, vol.38, no.11, 2005, pp. 2007-2021.

J3. **Jinwei Gu**, Jie Zhou, and David Zhang. "A Combination Model for Orientation Field of Fingerprints". *Pattern Recognition*, vol. 37, no.3, 2004, pp. 543-553.

J2. Jie Zhou and **Jinwei Gu**, "A Model-based Method for the Computation of Fingerprints' Orientation Field". *IEEE Transactions on Image Processing,* vol. 13, no. 6, 2004, pp 821-835.

J1. Jie Zhou and **Jinwei Gu**, “Modeling Orientation Fields of Fingerprints with Rational Complex Functions". *Pattern Recognition,* vol. 37, no. 2, 2004, pp 389-391.

Refereed conference proceedings:

C60. Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, Jinwei Gu. "Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning". ICLR 2026.
C61. Xiao Fu, Shitao Tang, Min Shi, Xian Liu, Jinwei Gu, Ming-Yu Liu, Dahua Lin, Chen-Hsuan Lin. "Plenoptic Video Generation". CVPR 2026.
C59. Lingen Li, Guangzhi Wang, Xiaoyu Li, Tianfan Xue, Ying Shan, Jinwei Gu. "CubeComposer: Spatio-Temporal Autoregressive 4K 360° Video Generation from Perspective Video", CVPR 2026.

C58. Mingde Yao, Menglu Wang, King Man Tam, Lingen Li, Tianfan Xue, **Jinwei Gu**. “PolarFree: Polarization-based Reflection-Free Imaging”, CVPR 2025.

C57. Zeqi Gu, Yin Cui, Max Li, Fangyin Wei, Yunhao Ge, **Jinwei Gu**, Ming-Yu Liu, Abe Davis, Yifan Ding. “ArtiScene: Language-Driven Artistic 3D Scene Generation Through Image Intermedlary”, CVPR 2025.

C56. Liqun Chen, Yuxuan Li, Jun Dai, **Jinwei Gu**, Tianfan Xue. “A Physics-Informed Blur Learning Framework for Imaging Systems”. CVPR 2025.

C55. Lingen Li, Zhaoyang Zhang, Yaowei Li, Jiale Xu, Xiaoyu Li, Wenbo Hu, Weihao Cheng, **Jinwei Gu**, Tianfan Xue, Ying Shan. “NVComposer: Boosting Generative Novel View Synthesis with Multiple Sparse and Unposed Images”, CVPR 2025.

C54. Hongjun Wang, Wonmin Byeon, Jiarui Xu, **Jinwei Gu**, Ka Chun Cheung, Xiaolong Wang, Kai Han, Jan Kautz, Sifei Liu. “Parallel Sequence Modeling via Generalized Spatial Propagation Network”, CVPR 2025.

C53. Yujin Wang, Tianyi Xu, Zhang Fan, Tianfan Xue, **Jinwei Gu**. “AdaptiveISP: Learning an adaptive image signal processor for object detection”, NeurIPS 2024.

C52. Xin Cai, Zhiyuan You, Hailong Zhang, **Jinwei Gu**, Wentao Liu, Tianfan Xue. “Phocolens: Photorealistic and consistent reconstruction in lensless imaging”, NeurIPS 2024.

C51. Ziran Zhang, Yongrui Ma, Yueting Chen, Feng Zhang, **Jinwei Gu**, Tianfan Xue and Shi Guo. “From Sim-to-Real: Toward General Event-based Low-light Frame Interpolation with Per-scene Optimization”, SIGGRAPH Asia 2024.

C50. Yongrui Ma, Shi Guo, Yutian Chen, Tianfan Xue, **Jinwei Gu**. “TimeLens-XL: Real-Time Event-Based Video Frame Interpolation with Large Motion”, ECCV 2024.

C49. Ruikang Li, Yujin Wang, Shiqi Chen, Fan Zhang, **Jinwei Gu**, Tianfan Xue. “DualDN: Dual-domain denoising via differentiable ISP”, ECCV 2024.

C48. Yutian Chen, Shi Guo, Fangzheng Yu, Feng Zhang, **Jinwei Gu**, Tianfan Xue. “Event-based Motion Magnification”, ECCV 2024.

C47. Yitong Jiang, Zhaoyang Zhang, Tianfan Xue, **Jinwei Gu**. “AutoDIR: Automatic all-in-one image restoration with latent diffusion”, ECCV 2024.

C46. ZhixiangWang, Baiang Li, Jian Wang, Yu-Lun Liu, **Jinwei Gu**, Yung-Yu Chuang, Shinichi Satoh. “Matting by Generation”, SIGGRAPH 2024.

C45. Zhaoyang Zhang, Wenqi Shao, Yixiao Ge, Xiaogang Wang, **Jinwe Gu**, Ping Luo. “Cached transformers: improving transformers with differentiable memory cache”, AAAI 2024.

C44. Gangwei Xu, Yujin Wang, **Jinwei Gu**, Tianfan Xue, Xin Yang, “HDRFlow: Real-time HDR Video Reconstruction with Large Motion”, CVPR 2024.

C43. Kechun Liu, Yitong Jiang, Inchang Choi, and **Jinwei Gu**. “Learning image-adaptive codebooks for class-agnostic image restoration”, ICCV 2023.

C42. Zhaoyang Zhang, Yitong Jiang, Wenqi Shao, Xiaogang Wang, Ping Luo, Kaimo Lin, **Jinwei Gu**. “Real-time Controllable Denoising for Image and Video”, CVPR 2023.

C41. Ruicheng Feng, Chongyi Li, Huaijin Chen, Shuai Li, **Jinwei Gu**, Chen Change Loy. “Generating Aligned Pseudo-Supervision from Non-Aligned Data for Image Restoration in Under-Display Camera”, CVPR 2023.

C40. Man Zhou, Yu Hu, Jie Huang, Feng Zhao, **Jinwei Gu**, Chen Change Loy, Deyu Meng, Chongyi Li. “Deep Fourier Up-Sampling”, *NeurIPS* 2022.

C39. Zhaoyang Zhang, Yitong Jiang, Jun Jiang, Xiaogang Wang, Ping Luo and **Jinwei Gu**. “STAR: Structure-Aware Lightweight Transformer for Real-time Image Enhancement”. ICCV 2021.

C38. Ke Yu, Zexian Li, Yue Peng, Chen Change Loy, and **Jinwei Gu**. “ReconfigISP: Reconfigurable Camera Image Processing Pipeline”. ICCV 2021.

C37. Zhaoyang Zhang, Wenqi Shao, **Jinwei Gu**, Xiaogang Wang, and Ping Luo. “Dynamic Quantization with Mixed Precision and Adaptive Resolution”. ICML 2021.

C36. Ruicheng Feng, Chongyi Li, Huaijin Chen, Shuai Li, Chen Change Loy, and **Jinwei Gu**. “Removing Diffraction Image Artifacts in Under-Display Camera via Dynamic Skip Connection Network.” *CVPR 2021*

C35. Kelvin Chan, Xintao Wang, Xiangyu Xu, **Jinwei Gu**, and Chen Change Loy. “GLEAN: Generative Latent Bank for Large-Factor Image Super-Resolution.” *CVPR 2021 (Oral)*

C34. Matthias Innman, Kihwan Kim, **Jinwei Gu**, Matthias Niesner, Charles Loop, Marc Stamminger, and Jan Kautz. *“NRMVS: Non-Rigid Multi-View Stereo”,* IEEE WACV 2020.

C33. Yan Chen, Jimmy Ren, Xuanye Cheng, Keyuan Qian, and **Jinwei Gu**. “*Very Power Efficient Neural Time-of-Flight*”, IEEE WACV 2020.

C32. Wei-Sheng Lai, Orazio Gallo, **Jinwei Gu**, Deqing Sun, Ming-hsuan Yang, and Jan Kautz. “Video Stitching for Linear Camera Arrays”, *BMVC 2019*.

C31. Soumyadip Sengupta, **Jinwei Gu**, Kihwan Kim, Guilin Liu, David Jacob, and Jan Kautz. *“Neural Inverse Rendering of an Indoor Scene from a Single Image”*, ICCV 2019.

C30. Chao Liu, **Jinwei Gu**, Kihwan Kim, Srinivas Narasimhan, and Jan Kautz. *“Neural RGB->D Sensing: Depth and Uncertainty from a Video Camera”*, CVPR 2019 (Oral) [**Best Paper Finalist**]

C29. Chen Liu, Kihwan Kim, **Jinwei Gu**, Yasutaka Funakawa, and Jan Kautz. *“PlaneRCNN: 3D Plane Detection and Reconstruction from a Single Image”*, CVPR 2019 (Oral)

C28. Donghoon Lee, Sifei Liu, **Jinwei Gu**, Mingyu Liu, Minghsuan Yang, Jan Kautz. “*Context-aware Synthesis and Placement of Object Instances*”. NIPS 2018.

C27. Sifei Liu, Guangyu Zhong, Shalini Gupta, **Jinwei Gu,** Varun Jampani, Ming-Hsuan Yang, and Jan Kautz. “*Switchable Temporal Propagation Network*”. ECCV 2018.

C26. Samarth Brahmbhatt*, **Jinwei Gu**, Kihwan Kim, James Hayes, and Jan Kautz. “*MapNet: Geometry-Aware Learning of Maps for Camera Localization*”. CVPR 2018 (Spotlight).

C25. Huaijin Chen*, **Jinwei Gu**, Orazio Gallo, Ming-Yu Liu, Ashok Veeraraghavan, and Jan Kautz. "*Reblur2Deblur: Video Deblurring via Self-Supervised Learning*". ICCP 2018.

C24. Patrick Wieschollek*, Orazio Gallo, **Jinwei Gu**, and Jan Kautz. "*Separating Reflection and Transmission Images in the Wild*". ECCV 2018.

C23. Sifei Liu*, Shalini Gupta, **Jinwei Gu**, Ming-Hsuan Yang, Jan Kautz. "*Learning Affinity via Spatial Propagation Networks.*" *NIPS, 2017*.

C22. Kihwan Kim, **Jinwei Gu**, Stephen Tyree, Pavlo Molchanov, Matthias Niebster, and Jan Kautz*. "A Lightweight Approach for Reflectance Estimation on-the-fly". ICCV 2017 (Oral)*

C21. Suren Jayasuriya*, Orazio Gallo, **Jinwei Gu**, Timo Aila, and Jan Kautz. *"Reconstructing Intensity Images from Binary Spatial Gradient Cameras", CVPR 2017 Embedded Vision Workshop (Oral)*

C20. Zhaopeng Cui*, **Jinwei Gu**, Boxin Shi, Ping Tan, and Jan Kautz*. "Polarimetric Multi-View Stereo". CVPR 2017*

C19. **Jinwei Gu**, Xiaodong Yang, Shalini Gupta and Jan Kautz*. "Dynamic Facial Analysis: From Bayesian Filtering to Recurrent Neural Networks". CVPR 2017*

C18. Suren Jayasuriya*, Orazio Gallo, **Jinwei Gu**, and Jan Kautz. *"Deep Learning with Energy-efficient Binary Gradient Cameras", arXiv preprint: 1612.00986.*

C17. Wei Jiang and **Jinwei Gu**. " Video Stitching with Spatial-Temporal Content-Preserving Warping". *IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Computational Cameras and Displays Workshop,* 2015.

C16. Chao Liu*, Gefei Yang*, and **Jinwei Gu**. "Learning Discriminative Illumination and Filters for Raw Material Classification with Optimal Projections of Bidirectional Texture Functions". *IEEE Conference on Computer Vision and Pattern Recognition (CVPR),* 2013.

C15. Jun Jiang*, Dengyu Liu*, **Jinwei Gu** and Sabine Susstrunk. "What is the Space of Spectral Sensitivity Functions for Digital Color Cameras?" *IEEE Workshop on the Applications of Computer Vision (WACV)*, 2013.

C14. Jun Jiang* and **Jinwei Gu**. "An Exemplar-Based Method for Automatic Visual Editing and Retouching of Fine Art Reproduction" *Color and Imaging Conference (CIC),* 2013.

C13. **Jinwei Gu** and Chao Liu*. "Discriminative Illumination: Per-Pixel Classification of

Raw Materials based on Optimal Projections of Spectral BRDFs". *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012 [**Oral**]

C12. Jun Jiang* and **Jinwei Gu**. "Recovering Spectral Reflectance under Commonly Available Lighting Conditions*". IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Computational Camera and Display Workshop*, 2012. [**Oral**]

C11. Adria Fores*, James Ferwerda, and **Jinwei Gu**. "Toward a Perceptually Based Metric for BRDF Modeling", *Proceedings IS&T/SID 20th Color Imaging Conference, November*, 2012.

C10. **Jinwei Gu,** Toshihiro Kobayashi*, Mohit Gupta, and Shree Nayar. "Multiplexed Illumination for Scene Recovery in the Presence of Global Illumination". *IEEE International Conference on Computer Vision (ICCV)*, 2011 [**Oral**]

C9. Yasunobu Hitomi*, **Jinwei Gu**, Mohit Gupta, Tomoo Mitsunaga, and Shree Nayar. "Video From a Single Exposure Coded Photograph using a Learned Over-Complete Dictionary". *IEEE International Conference on Computer Vision (ICCV)*, 2011.

C8. **Jinwei Gu**, Yasunobu Hitomi, Tomoo Mitsunaga, and Shree Nayar. "Coded Rolling Shutter Photography: Flexible Space-Time Sampling". *IEEE International Conference on Computational Photography (ICCP),* Mar. 2010. [**Oral**]

C7. **Jinwei Gu**, Shree Nayar, Eitan Grinspun, Peter Belhumeur, and Ravi Ramamoorthi. "Compressive Structured Light for Recovering Inhomogeneous Participating Media". *European Conference on Computer Vision (ECCV),* Oct. 2008, pp 845-858. [**Oral**]

C6. **Jinwei Gu**, Ravi Ramamoorthi, Peter Belhumuer, and Shree Nayar. "Dirty Glass: Modeling and Rendering Contaminations on Transparent Surfaces". *Eurographics Symposium on Rendering (EGSR),* June 2007. [**Oral**]

C5. Jie Zhou, **Jinwei Gu,** and David Zhang, "Singular Points Analysis in Fingerprints Based on Topological Structure and Orientation Field". *International Conference of Biometrics (ICB),* August 2007, pp 261-270.

C4. Stephen Lin, **Jinwei Gu**, Shuntaro Yamazaki, and Heung-Yeung Shum. "Radiometric Calibration from a Single Image". *IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR),* June 2004, vol. 2, pp 938-945. [**Oral**]

C3. Wei Jiang, Mingjing Li, Hongjiang Zhang, and **Jinwei Gu**. "Online Feature Selection based on Generalized Feature Contrast Model". *IEEE International Conference on Multimedia and Expo (ICME), 2004*.

C2. **Jinwei Gu** and Jie Zhou. "Model-based Orientation Field Estimation for Fingerprint Recognition". *IEEE International Conference on Image Processing (ICIP)*, 2003.

C1. **Jinwei Gu** and Jie Zhou, "A Novel Model for Orientation Field of Fingerprints". *IEEE Computer Society Conference on Computer Vision and Pattern Recognition*

(CVPR), June 2003, vol 2, pp 493-498.

Preprints and Arxiv:

A4. NVIDIA. “Cosmos World Foundation Model Platform for Physical AI”, Arxiv 2501.03575.

A5. NVIDIA. "Cosmos-Reason1: From Physical Common Sense To Embodied Reasoning." Arxiv 2503.15558. 2025.

A6. NVIDIA. "Cosmos-Transfer1: Conditional World Generation with Adaptive Multimodal Control." Arxiv 2503.14492. 2025.

A7. NVIDIA. "World Simulation with Video Foundation Models for Physical AI (Cosmos-Predict2.5 / Cosmos-Transfer2.5)." Arxiv 2511.00062. 2025.

A8. Wei-Cheng Tseng, Jinwei Gu, Qinsheng Zhang, Hanzi Mao, Ming-Yu Liu, Florian Shkurti, Lin Yen-Chen. "Scalable Policy Evaluation with Video World Models." Arxiv 2511.11520. 2025.

A9. Shenyuan Gao, William Liang, Kaiyuan Zheng, ..., Jinwei Gu, et al. "DreamDojo: A Generalist Robot World Model from Large-Scale Human Videos." Arxiv 2602.06949. 2026.

A10. Xiao Fu, Shitao Tang, Min Shi, Xian Liu, Jinwei Gu, Ming-Yu Liu, Dahua Lin, Chen-Hsuan Lin. "Plenoptic Video Generation." Arxiv 2601.05239. 2026.

A3. J. Dai, L. Chen, X. Wang, Y. Hu, J. Gu, T. Xue. Tolerance-Aware Deep Optics. Arxiv 2502.04719.

A2. L. Li, M. Yao, X. Meng, M. Yu, T. Xue, J. Gu. “Uni-ISP: Unifying the Learning of ISPs from Multiple Cameras”, arXiv 2406.01003.

A1. Yitong Jiang, Inchang Choi, Jun Jiang, and **Jinwei Gu**. “HDR Video Reconstruction with Tri-Exposure Quad-Bayer Sensors”. ArXiv. 2021.

Book Chapters, monographs and tutorials:

Jie Zhou, David Zhang, **Jinwei Gu**, and Nannan Wu. “Graphical Representation of Fingerprint Images". Integrated Image and Graphics Technologies, Kluwer academic publisher, Boston, 2004.

Theses

“Measuring, Modeling, and Synthesis of Time-Varying Appearance of Natural Phenomena”, Columbia University, May 2010. (PhD Thesis)

“Fingerprint Orientation Field Modeling and Its Applications”, Tsinghua University, May 2005. (MS Thesis)

Popular press:

Dynamic Facial Analysis with RNNs, GTC 2017 (GPU Technology Conference) Invited Talk

Polarimetric Multiview Stereo. ICCP 2018 Poster Presentation.

Space-Time Appearance Factorization, Computer Graphics World, Aug 2006.

PATENTS:

Patent Granted:

P20. **Jinwei Gu**, Kihwan Kim, Jan Kautz, Guilin Liu, and S. Sengupta. Inverse Rendering of a Scene from a Single Image. US Patent 11295514B2.

P19. Kihwan Kim, **Jinwei Gu**, Chen Liu, and Jan Kautz. 3D Plane Detection and Reconstruction Using a Monocular Image. US Patent 11037051B2.

P18. **Jinwei Gu**, Kihwan Kim, and Chao Liu. Estimating Depth for a Video Stream Captured with a Monocular RGB Camera. US Patent 10984545.

P17. Sifei Liu, Shalini De Mello, **Jinwei Gu**, Varun Jampani, and Jan Kautz. Switchable Propagation Neural Network. US Patent 11328173B2

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